Università Bocconi
Master of Science in Data Science and Business Analytics

Project Report for the course 20592 – Statistics and Probability

Computational Statistics - Final Project

Academic Year	2025 / 2026
Course Instructors	Prof. Sonia Petrone Dr. Martin Chak Dr. Louise Alamichel
Students	Erdem Bayrakcioglu – 3353141 Henricus Bracht – 3237014 Leonard Fischbach – 3374531
Submitted in	Milan, January 10th, 2026

Table of Contents

1 Introduction

This paper deals with the analysis of two datasets using both a Fisher scoring algorithm as well as a Markov Chain Monte Carlo method. The first dataset "CreditCard" will be analyzed using Logistic regression to determine which covariates influence the receipt of a credit card (binary outcome "accepted" vs. "not accepted" and how. For the second dataset "azcabgptca" (herafter referred to as "hospital"), we model the count outcome "los" with a Poisson regression which measures patients' length of stay in the hospital using patient and procedure information.
We then complement maximum-likelihood estimation with a Bayesian analysis using Markov Chain Monte Carlo (MCMC), including convergence diagnostics and interpretation of key covariate effects. Find exercise 1b)'s proof and the plots concerning sections 3.1 and 3.2 in the appendix.

2 Generalized Linear Models and Estimation Algorithms

2.1 Fisher Scoring Algorithm and its Relevance for GLMs

The Fisher scoring algorithm is an iterative optimization method used to estimate the parameters of Generalized Linear Models (GLMs) by maximizing the log-likelihood function. Unlike classical linear regression, GLMs allow the response variable to follow a distribution from the exponential family, such as the Bernoulli or Poisson distribution, and relate the conditional mean of the response to a linear predictor through a link function.

In a GLM, the systematic component is given by the linear predictor $\eta = X\beta$, where X denotes the design matrix and \beta is the vector of regression coefficients. The expected value of the response variable is connected to the linear predictor via a link function. In this project, canonical link functions are used, namely the logit link for logistic regression and the log link for Poisson regression.

Fisher scoring is closely related to the Newton–Raphson optimization method. While Newton's method relies on the observed Hessian of the log-likelihood, Fisher scoring replaces it with its expected value, known as the Fisher information matrix. This substitution often leads to improved numerical stability and makes Fisher scoring particularly suitable for parameter estimation in GLMs, especially when canonical links are employed.

The algorithm updates the parameter vector iteratively until convergence, yielding maximum likelihood estimates of the regression coefficients. These estimates can be directly compared to those obtained from built-in GLM implementations, which provides a useful validation of the custom Fisher scoring implementation.

2.2 Analytical Derivation of the Fisher Scoring Quantities

This section derives the quantities used in the Fisher scoring algorithm. The derivations apply to the Logistic and Poisson regression models implemented in this project.

Let $Y_1, \ldots, Y_n$ be independent observations from an exponential family distribution with conditional mean μ_i. For GLMs with canonical link functions, the natural parameter equals the linear predictor, i.e. $\theta_i = \eta_i = X_i^T \beta$. The log-likelihood can therefore be written as:

$$\mathrm{l}(\beta) = \sum_{i=1}^{n} (y_i \theta_i - b(\theta_i)) + \text{const}$$

The score function, defined as the slope of the log-likelihood with respect to β, is given by:

$$\mathrm{U}(\beta) = \sum_{i=1}^{n} x_i (y_i - \mu_i) = X^T (y - \mu)$$

The Fisher scoring algorithm uses the Fisher information matrix, defined as the negative expected value of the second derivative of the log-likelihood:

$$\mathrm{I}(\beta) = -\mathrm{E}\left[\frac{\partial^2 \mathrm{l}(\beta)}{\partial \beta \ \partial \beta^\top}\right]$$

For GLMs with canonical links, this matrix simplifies to $\mathrm{I}(\beta) = \mathrm{X}^\top \mathrm{W}\,\mathrm{X}$, where W is a diagonal weight matrix. In Logistic regression with Bernoulli responses, the variance function is $\mu_i (1 - \mu_i)$, resulting in weights $\mathrm{W_i} = \mu_i (1 - \mu_i)$. In Poisson regression, the variance equals the mean, yielding $W_i = \mu_i$.

With the score vector and the Fisher information matrix, we can formulate the Fisher scoring update equation as follows: $\beta^{(k+1)} = \beta^{(k)} + \mathrm{I}\left(\beta^{(k)}\right)^{-1} \mathrm{U}\left(\beta^{(k)}\right)$.

In the implementation, this update is computed by solving a linear system involving $I(\beta)$ and $U(\beta)$. The algorithm iterates until convergence (or maximum iterations reached), producing maximum likelihood estimates of the model parameters.

2.3 Canonical versus Non-Canonical Links in Fisher Scoring

In a linear regression, we have $Y = X^\top B + \epsilon$ with $E[Y \mid X] = X^\top B$, where Y is our dependent variable, X is our covariates and B are our parameters and ϵ is an error term. However, we face one issue with Generalized Linear Models: The support of Y could be binary (logistic regression) or counting values (Poisson), therefore Y is not normally distributed and $E[Y \mid X]$ is restricted such that $\mu \neq X^\top B$ (in general). To be able to maintain a linear structure in our covariates, we can use a link function $g(\mu) = X^\top B$.

For the canonical link function, we define $g(\mu) = (b')^{-1}(\mu)$ and we now have $g(\mu) = X^\top B = \theta$, since $\mu = b'(\theta)$. As shown in the proof, our score function (first gradient) and in the second order gradient, the y vanishes because it is linear in the canonical parameter and the Hessian thus does not depend on y anymore.

For non-canonical links this equality $g(\mu) = (b')^{-1}(\mu)$ does not hold anymore, thus $\theta_i = (b')^{-1}\left(g^{-1}(X_i^\top B)\right)$, thus if we plug that into the log-likelihood function, when calculate the derivative, as θ_i is not linear in B, the chain rule produces further terms such that it still contains the y_i and therefore the Hessian still depends on the data. This implies that the $E[\nabla^2 \ell(B)] \neq \nabla^2 \ell(B)$ (in general).

3 Bayesian Inference and MCMC methodology

3.1 MCMC Algorithm, Acceptance Probabilities and Prior Distributions

In Bayesian GLMs the likelihood is the same as in the frequentist GLM, but the coefficients are assigned a prior distribution and inference is based on the posterior. For logistic and Poisson regression this posterior has no closed-form expression, so we rely on MCMC. MCMC is essential here because it produces draws from the posterior, enabling uncertainty quantification via credible intervals and supporting interpretation of covariate effects. We implement a Random-Walk Metropolis–Hastings sampler, discard burn-in, and summarize the posterior with means and credible intervals. Convergence and mixing are assessed using trace plots and autocorrelation for key parameters (income and procedure). We assign independent Gaussian priors to all coefficients,

$$\beta_j \sim N(0, prior\_sd^2).$$

This weakly informative prior shrinks effects toward zero (stabilizing estimation and preventing extreme coefficients) while still allowing the data to dominate, especially since predictors are standardized.

3.2 Convergence Diagnostics

We assess convergence and mixing of the Random-Walk Metropolis–Hastings chains using standard MCMC diagnostics. First, we inspect trace plots for the coefficients of income (logistic model) and procedure (Poisson model). In both cases, the traces fluctuate around a stable level after burn-in and show no visible trend, which is consistent with stationarity. Second, we report the autocorrelation function (ACF) up to lag 50 for the same parameters. The ACF decreases with the lag, indicating dependence typical of random-walk samplers but still sufficient mixing given the long post–burn-in chains. Together with the acceptance rates, these diagnostics support that the retained draws provide a reasonable approximation of the posterior.

4 Application to the Datasets and Results

4.1 CreditCard Dataset

The McFadden pseudo R^2 (0.30 rounded both Fisher and MCMC) demonstrates that that the model explains the data quite well. Among the covariates, by far the most important is the *reports* covariate with a value of -2.356 (Fisher) / -2.4 (MCMC) showing that an increase in negative credit reports has a major effect as it decreases the odd of getting a credit card by 90.5% (Fisher) / 90.9% (MCMC) which also makes intuitively sense as these credit score indicate bad creditworthiness. The second most important variable is the *active* with 0.834 (Fisher) / 0.849 (MCMC), more specifically, the number of active credit accounts. This indicates that having additional credit accounts already positively influences the odd of getting a credit card by c. 130% (Fisher) / 134% (MCMC) which is an interesting finding as there might be logical reasons that this could negatively influence credit risk. By absolute values, being self-employed (-0.757 Fisher / -0.726 MCMC) has a strong negative effect. However, this is a dummy variable and the majorcards is continuous and the effect is thus more dramatic (0.505 Fisher / 0.500 MCMC), increasing the odd by 65% (both Fisher and MCMC) per other credit card for getting a credit card.

4.2 Hospital Dataset

We excluded the "died" variable as it is an endogenous variable to the treatment and could hence falsify our results.

The McFadden pseudo R^2 (both Fisher / MCMC rounded 0.28) demonstrates that we have a good fit with this Poisson regression. For the dependent variable length of stay, the three most influential covariates are procedure, age and admission type (listed by descending importance). Procedure is either CABG or PTCA where CABG is more invasive as PTCA, thus it makes sense that someone taking the more invasive treatment is on average staying c. 3.1x (for both Fisher/MCMC) longer in the hospital. Per increasing year of age, a patient stays on average 4% (for both Fisher/MCMC) longer in the hospital and if the treatment is urgent a patient stays 21% (for both Fisher/MCMC) longer on average. We consider age to be more important as being just more than c. 4.5 years older means that it has a bigger effect on the length of the stay than whether the procedure is urgent or not.

Overall, Fisher Scoring and MCMC yield almost the same result as you can easily see in the appendix.

5 Appendix

Appendix 1 – Newton's method and Fisher scoring coincide under canonical link functions

We have Newton's method from our lectures:

$$\theta_{t+1} = \theta_t - [\nabla^2 \ell(\theta_t)]^{-1} \nabla \ell(\theta_t)$$

And we have Fisher Scoring from our lectures:

$$\theta_{t+1} = \theta_t + I_n(\theta_t)^{-1} \nabla \ell(\theta_t)$$

For the exponential family, we are using the pdf from the a-b-c formula

$$f(y_i, \theta_i, \phi) = \exp\left\{\frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)\right\}, \text{ where } b'(\theta_i) = E(Y_i) = \mu_i$$

The likelihood function is $\mathcal{L}(\theta) = \prod_{i=1}^{n} f(y_i, \theta_i, \phi)$ and the log-likelihood function $\ell(\theta) = \log \mathcal{L}(\theta) = \sum_{i=1}^{n} \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)$.

Under the canonical link function we have $\eta_i = X_i^\top B = g(\mu_i)$ with $g(\mu_i) = b'^{-1}(\mu_i) = \eta_i = X_i^\top B = \theta_i$.

We can plug into our log-likelihood function now:

$$\ell_i(B) = \frac{y_i X_i^\top B - b(X_i^\top B)}{a(\phi)} + c(y_i, \phi)$$

Computing the first and second order gradient

$$\nabla \ell_i(B) = \frac{y_i X_i^\top - b'(X_i^\top B) X_i}{a(\phi)} \text{ and } \nabla^2 \ell_i(B) = \frac{-X_i X_i^\top b''(X_i^\top B)}{a(\phi)}$$

Notice how y_i vanishes when taking the second order gradient. Let $W_{ii} = b''(X_i B)$, then

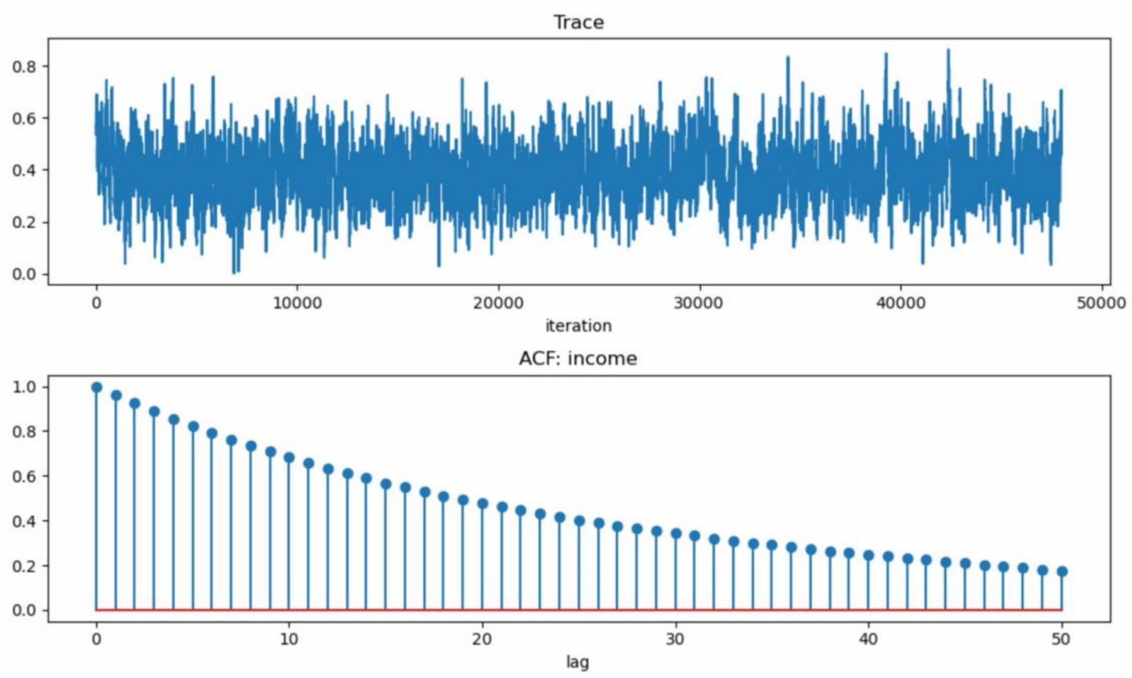
$$\nabla^2 \ell_i(B) = \frac{-X^\top W X}{a(\phi)}$$

Since y_i is now vanished and Y_i is the only random variable, the expected value is the value itself as the Hessian is no longer a function of a random variable.

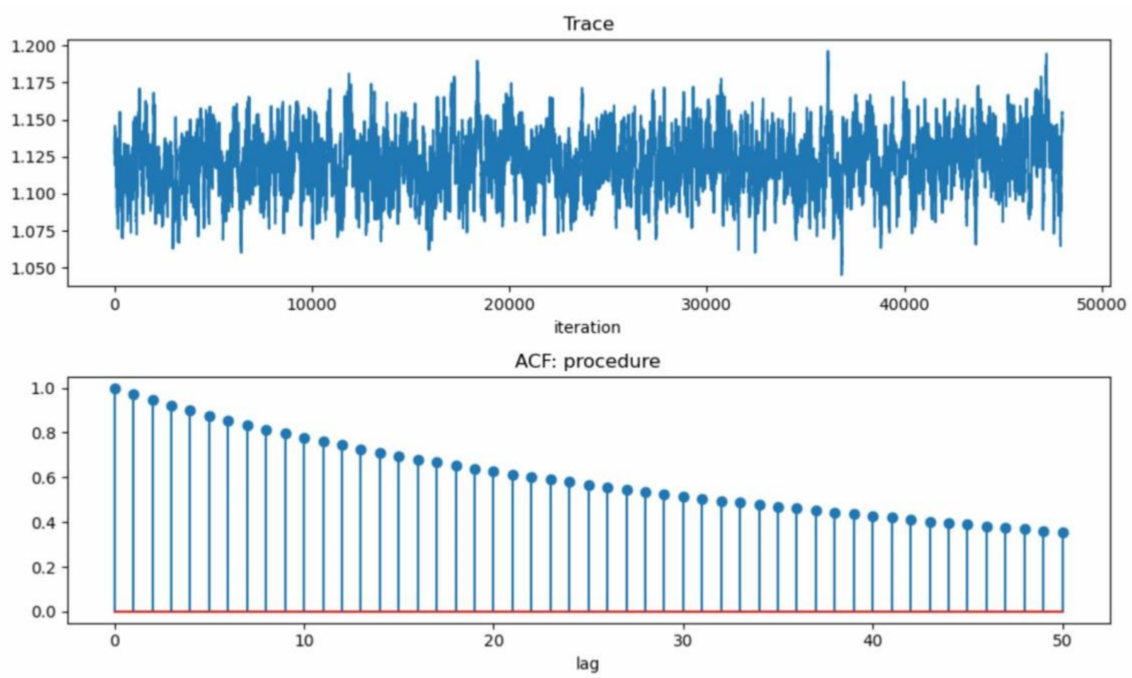
$$E[\nabla^2 \ell(B)] = \nabla^2 \ell(B)$$

From the book (Chapter 9.10) we are given $I_n(B) = -E_\theta[\nabla^2 \ell(B)]$. Under canonical links, therefore $I_n(B) = -E_\theta[\nabla^2 \ell(B)] = -\nabla^2 \ell(B)$. This shows that the two equations shown at the top are the same.

Appendix 2 – Trace Plots & Autocorrelation Function Plots: Credit Card Dataset



Appendix 3 – Trace Plots & Autocorrelation Function Plots: Hospital Dataset



Appendix 4 - Regression results Fisher scoring

Hospital Dataset (Poisson)		
Coeff. name	**Coeff value**	**Wald p-value**
intercept	1.284	0.0

age	0.041	0.0
procedure	1.122	0.0
gender	-0.103	0.0
type	0.190	0.0
Mc Fadden pseudo R^2	0.280	

CreditCard Dataset (Logistic)		
Coeff. name	**Coeff value**	**Wald p-value**
intercept	0.887	0.0
reports	-2.356	0.0
age	-0.127	0.192
income	0.383	0.0
dependents	-0.302	0.0
months	0.034	0.714
majorcards	0.505	0.0
active	0.834	0.008
owner	0.478	0.017
selfemp	-0.757	0.009
Mc Fadden pseudo R^2	0.302	

Appendix 5 - MCMC results

Hospital Dataset	
Coeff. name	**Coeff value**
intercept	1.285
age	0.041
procedure	1.121
Gender	-0.103
type	0.190
McFadden pseude R^2	0.280

CreditCard Dataset	
Coeff. name	**Coeff value**
intercept	0.902

reports	-2.400
age	-0.128
income	0.389
dependents	-0.304
months	0.040
majorcards	0.500
active	0.849
owner	0.483
selfemp	-0.726
Mc Fadden pseudo R^2	0.302